

SET-A

Scholar Number: _____

Name: _____

Maulana Azad National Institute of Technology Bhopal

Department of Electrical Engineering

MINI TEST EXAMINATION

EE-214 MEASUREMENT AND INSTRUMENTATION

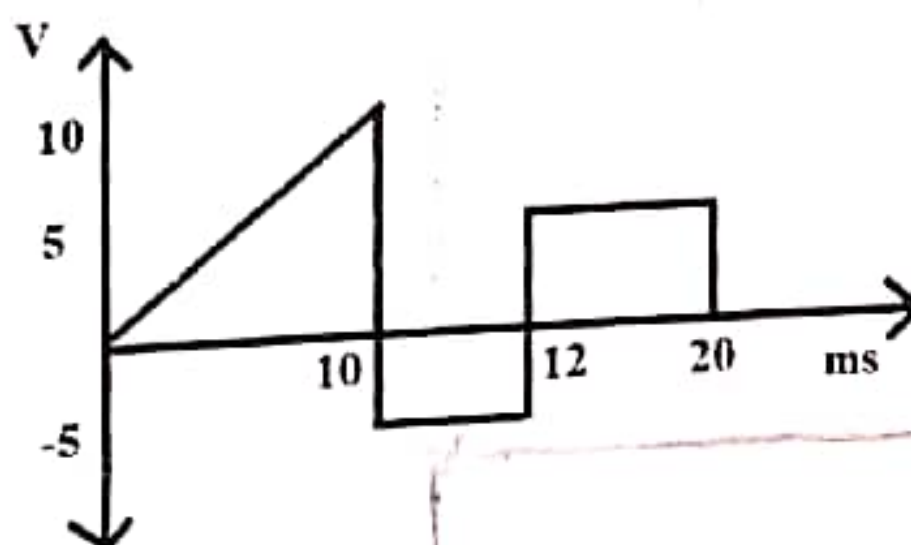
Subject Coordinator Names: Dr. Sushma Gupta & Dr. More Raju

Max Marks: 25

Time: 1 Hour

Date: 24/08/2023

1. Differentiate between mechanical, electrical and electronics instruments. [5 M]
2. What is the absolute type of instrument? Explain through the suitable example. [5 M]
3. Explain the active and passive type of transducer with suitable examples. [5 M]
4. A periodic time varying voltage is measured across a load by an oscilloscope having a waveform shown below. If a PMMC meter is connected across the same load, find its reading. [5 M]



5. The following data refers to a moving coil voltmeter.
Resistance = $10 \text{ K}\Omega$, Coil dimensions = $30\text{mm} \times 30\text{mm}$, Number of coil turns = 100, Air gap flux density = 0.08 Wb/m^2 , Spring constant = $3 \times 10^{-6} \text{ Nm/degree}$. Find the deflection produced for a voltage of 200 V. [5 M]